

TeleMedAI: An Intelligent Automated Platform for Tele-Medical Underwriting for Health Insurance Applications

Abstract

Health insurance underwriting frequently requires additional medical verification for policies that cannot be processed through straight-through processing (STP), necessitating detailed health assessments before a policy can be approved. Traditionally, this verification is carried out by tele-handlers who manually call customers, ask a fixed set of health questionnaires, probe further based on responses, and pass the collected information to underwriters for risk classification. This manual process is time-consuming, inconsistent across different handlers, difficult to scale during high policy volumes, and often results in delayed turnaround times for customers. Existing systems typically automate only isolated parts of this workflow, leaving data collection, response validation, and risk classification disconnected and reliant on manual effort. This project introduces an Intelligent Automated Platform for Tele-Medical Underwriting, which integrates a conversational bot to obtain customer consent and administer health questionnaires, dynamically determine whether additional drill-down questions are required based on prior responses, structure the collected conversational data into a standardized digital format, and pass it through a rule-based decisioning engine to classify each case for automated approval or manual review. The platform further visualizes the underwriting journey and generates structured case summaries through an interactive interface for underwriters. The objective of this project is not to eliminate human oversight in underwriting, but to automate routine tele-medical data collection and classification, reduce the workload on tele-handlers and underwriters, and enable faster, more consistent, and scalable underwriting decisions.

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